

Symbols & Labeling

Diagram Symbols and Notation

	Water		Valve
	Dirty Water		Stop-Check Valve
	Air Low Pressure		Filter
	Air Ambient Pressure		Pressure/Vacuum Sensor
	Air High Pressure		Pipe
	Clothing Item		Pump

Flow Diagram Key

 Flow Arrow shows the direction of fluid flow through a pipe

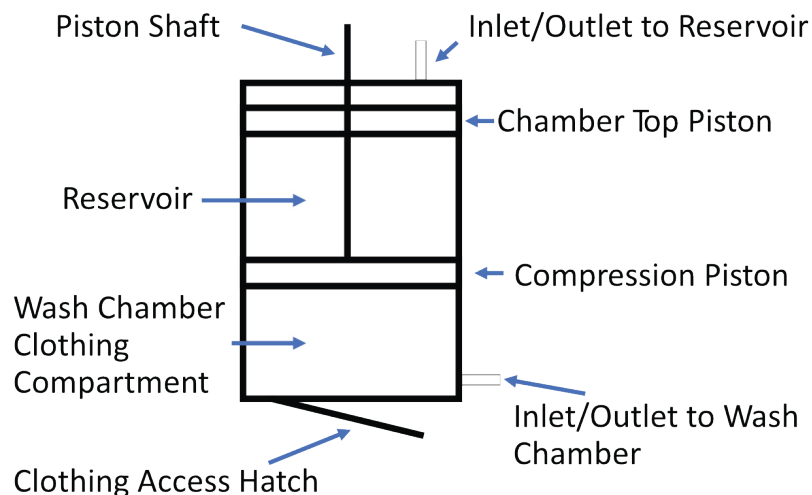
 Closed Valve

 Open Valve has fill color corresponding to the fluid passing through it

 Example of water flowing through open valve

 Example of closed valve blocking flow

Washing Machine Pipe Symbol



Washing Machine P&ID

Basic Washing Machine Cycle Piping Diagram

Labeled Components:

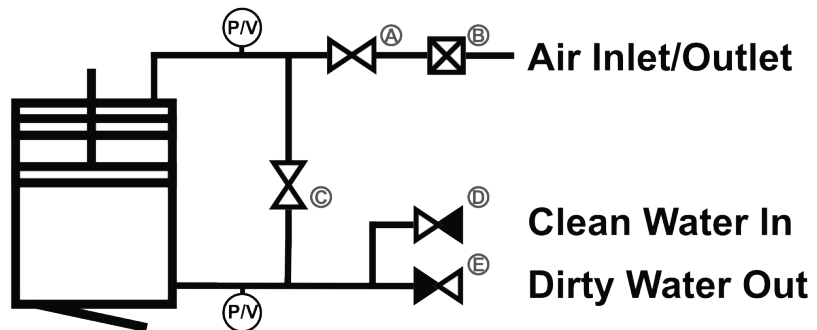
A: Air Valve

B: Water Trap/Sensor

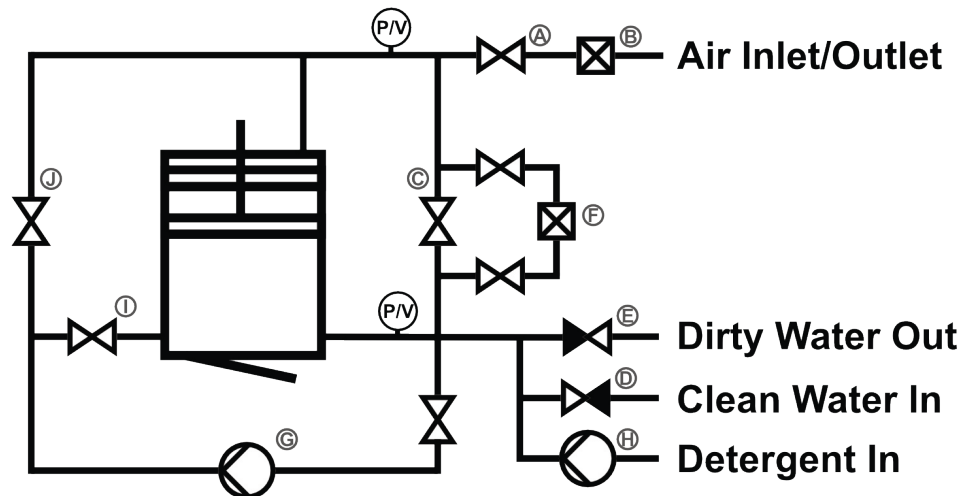
C: Isolation Valve

D: Clean Water Valve

E: Dirty Water Valve



Piping Diagram with Potential Improvements to Design Through Additional Design Spaces



Labeled Components:

F: Water Filter

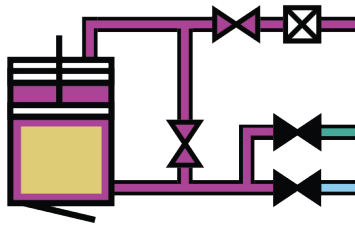
G: Water Circulation and Vacuum Pump

H: Detergent Dosing Pump

I: Water Circulation Valve

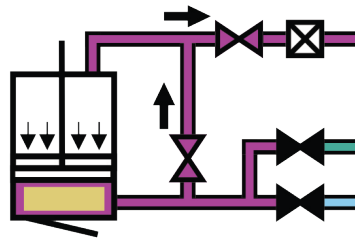
J: Vacuum Valve

Hydration Phase Flow Diagram

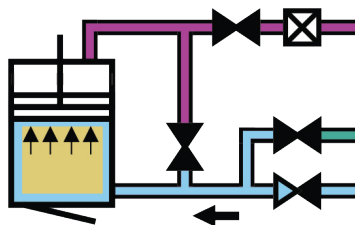


User inserts clothing item into fully dry wash chamber.

User initiates start of wash cycle with inputs such as clothing type and level of contaminant present.

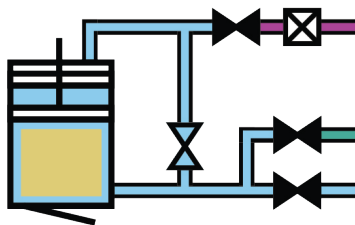


Isolation valve and air valve open to allow air to exit chamber as both compression piston and chamber top travel down to compress clothing item and displace air in the chamber.



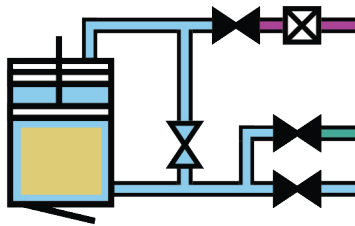
Isolation valve closed and water inlet open as pistons retract, drawing clean water into wash chamber.

Pistons stop when chamber volume matches a preset value declared by the user.

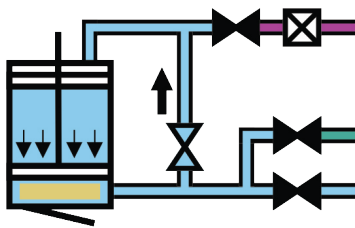


Wash chamber is flooded with water, and the washing machine is ready to begin agitation phase of wash cycle.

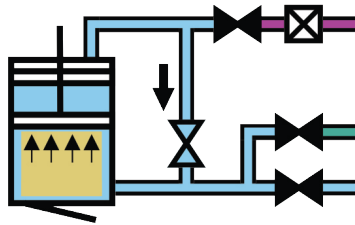
Agitation Phase Flow Diagram



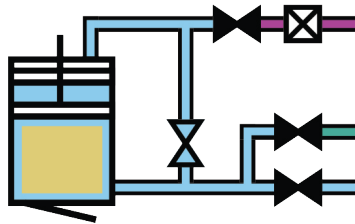
Wash chamber is fully flooded with water, and just finished hydration and soaking.



Clothing item is compressed with 50 Psi equivalent force. Water in wash chamber flows through clothing, agitating it and flows to chamber top reservoir.

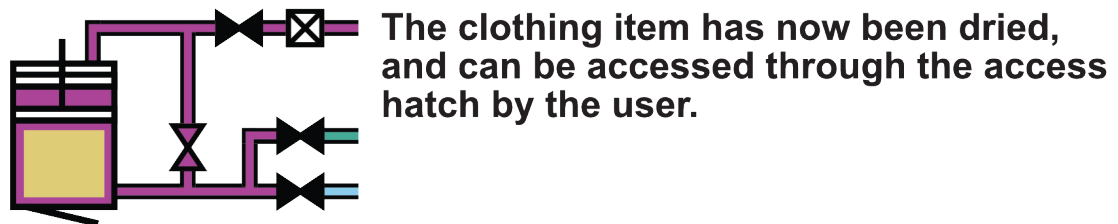
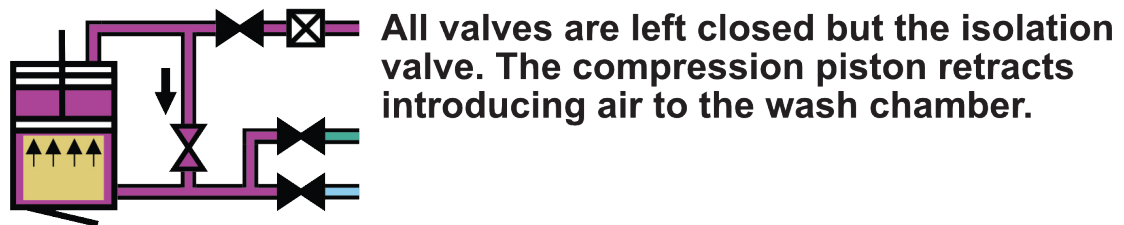
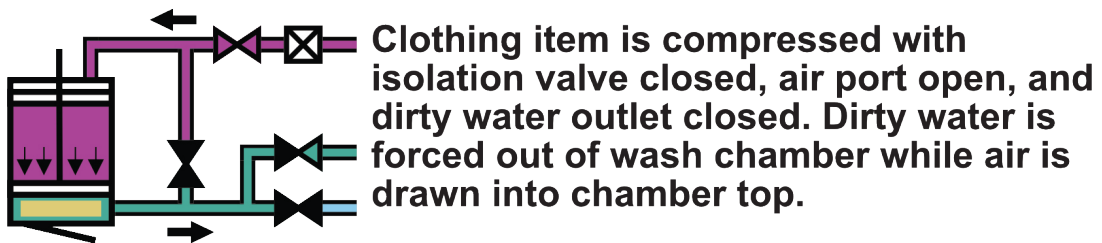
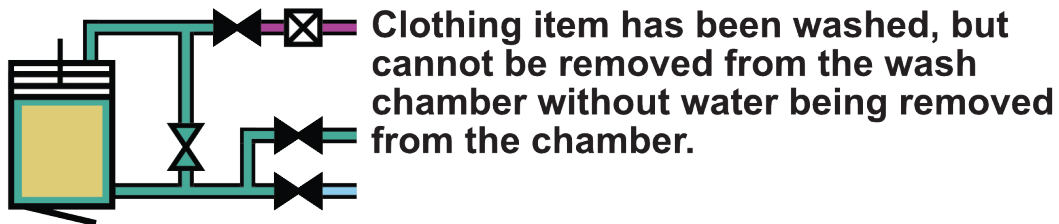


- **Compression piston retracts, forcing water from reservoir back to wash chamber.**

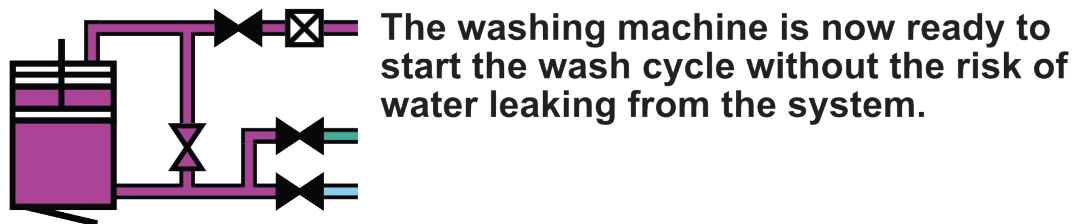
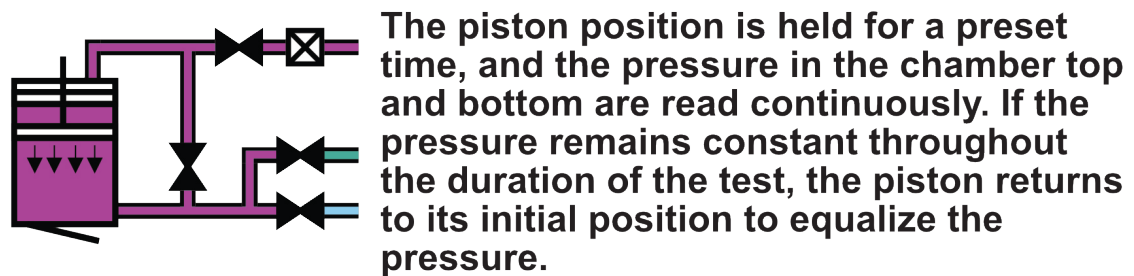
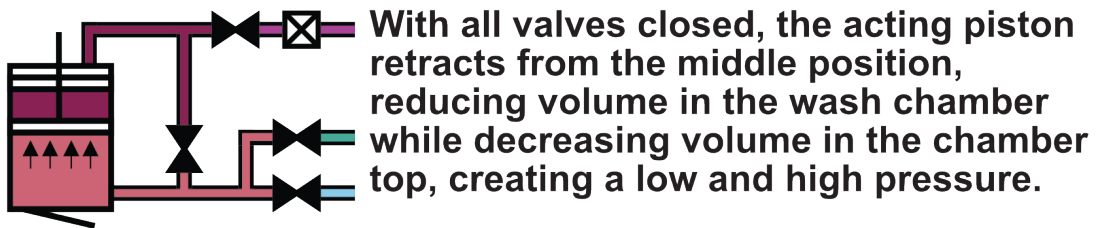


■ The washing machine is now in the same state it has started the compression cycle in. Either another stage of compression or removal of the dirty water can be accomplished from this state.

Removal of Dirty Water Phase Flow Diagram

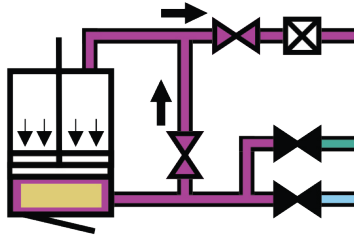


Pre-Wash Leak Check Flow Diagram

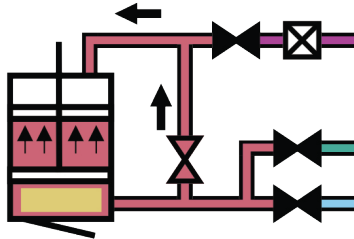


Flow Diagram for Drawing a Vacuum in Wash Chamber

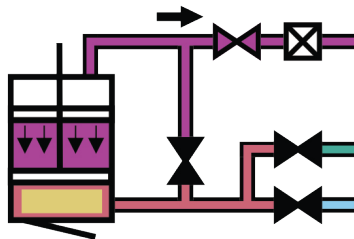
Note: Drawing vacuum in the wash chamber could potentially be used to aid in the hydration and drying phases of the wash cycle, allowing the machine to draw more air and water out of the wash chamber than compression alone could achieve. Pictured alone would be the process of drawing a vacuum prior to the hydration phase of the wash cycle.



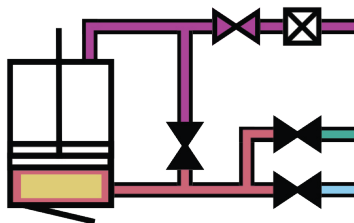
Compression of the clothing item has just been completed in the same way it is accomplished for the normal hydration phase of the wash process.



With all inlet/outlet valves closed, the chamber top piston retracts fully, drawing air from the wash chamber and increasing the volume in the chamber, lowering the pressure.



The isolation valve is closed, then the air inlet is opened. The chamber top pressure is allowed to equalize, and the chamber top piston is extended to meet the compression piston.



Each of the pistons are now in the same place as during the start of the cycle, only the wash chamber is at a lower pressure than when this process started. This can be repeated until the pressure inside the wash chamber is minimized.